



Linear Algebra for Computer Science

T-201-LINC

EXAM

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Date: April 25th, 2025

Time: 9:00 am - 12:00 pm

Name:

ID Number (Kennitala):

See next page for further instructions.

Instructions

Schedule for the exam: The exam starts at 9:00 am. You have until 12:00 pm to solve the problems of the exam.

The exam is in three parts:

- Part 1 Calculations. This part has 7 problems, but the **best 6 count**. Each is worth 8%.
- Part 2: Coding and Algorithms. 4 problems, but the **best 3 count**. Each is worth 12%.
- Part 3: Theory. This part has 3 problems, but the **best 2 counts**. Each is worth 8%.

Permitted exam materials:

- A single sheet of A4 paper. You can write on the front and the back. The paper is (1) handwritten by pen and/or (2) digitally handwritten and printed out. (3) Text typed e.g., via keyboard is also possible but with a minimum font size 10 with font Times New Roman / Arial (or another font of roughly the same size with font size 10). First and last name needs to be at the top of the paper. The paper may not contain printed out lecture slides (or screenshots of slides and other content found online or from digital sources - this includes screenshots of proofs, figures and graphs).
- A calculator. Not a laptop, a tablet, a smartphone, or a device that connects to the internet. The calculator may not be a scientific or a graphical calculator.

You are not allowed to communicate with each other or get external help. Your solutions must be your own.

You should always show your work.

- If you use a formula make sure to write it in your solution, even though you use a calculator to calculate the value.
- Explain and motivate your work.
- When you need to write code in your solutions, we will not deduct points for simple syntax errors, and pseudo-code is acceptable. Note, however, that the grade for questions with code is determined both by correctness and running time.

Good luck!

Part 1

- (a) (4%) Consider the linear system given below. Solve the system, in any way you want.

$$-2x_1 + 3x_2 + 5x_3 = 2$$

$$2x_2 + 3x_3 = 5$$

$$3.5x_3 = -3.5$$

Show work.

☐ Need more space? Continued here: _____

$$\begin{array}{lll}
 1) & 3.5x_3 = -3.5 & 2) \quad 2x_2 + 3(-1) = 5 \\
 & \boxed{x_3 = -1} & 2x_2 - 3 = 5 \\
 & & 2x_2 = 8 \\
 & & \boxed{x_2 = 4} \\
 & & 3) \quad -2x_1 + 3(4) + 5(-1) = 2 \\
 & & -2x_1 + 12 - 5 = 2 \\
 & & -2x_1 + 7 = 2 \\
 & & -2x_1 = -5 \\
 & & \boxed{x_1 = \frac{5}{2}}
 \end{array}$$

$$\begin{bmatrix} 2.5 \\ 2.5 \\ 1 \\ -1 \end{bmatrix}$$

- (b) (4%) Solve the corresponding homogeneous linear system, in any way you want. What does the result imply about whether the solution in subtask (a) is unique? Why?

Show work.

☐ Need more space? Continued here: _____

homogeneous:

$$Ax = 0$$

$$\begin{array}{l}
 -2x_1 + 3x_2 + 5x_3 = 0 \\
 2x_2 + 3x_3 = 0 \\
 3.5x_3 = 0
 \end{array}$$

$$\begin{array}{l}
 \underline{x_3 = 0} \\
 2x_2 + 0 = 0 \\
 \downarrow \\
 \underline{x_2 = 0} \\
 -2x_1 + 0 + 0 = 0 \\
 \downarrow \\
 x_1 = 0
 \end{array}$$

We only got the zero vector as a solution to the homogeneous system, which means that the "all" solutions (one unique) to $Ax = b$ where

$$\begin{bmatrix} -2 & 3 & 5 \\ 0 & 2 & 3 \\ 0 & 0 & 3.5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \\ -3.5 \end{bmatrix}$$

is $\vec{x} = \vec{p} + \text{null}(A)$

$$= \begin{bmatrix} 2.5 \\ 4 \\ -1 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2.5 \\ 4 \\ -1 \end{bmatrix}, \text{ which is unique.}$$

4. (a) (4%) Below are three matrices, A , B and C .

$$A = \begin{pmatrix} 1 & -1 & 0 \\ 2 & 0 & -1 \\ -1 & 1 & 1 \end{pmatrix} \quad B = \begin{pmatrix} 0.5 & -0.5 \\ -1 & 1 \\ 1 & -1 \end{pmatrix} \quad C = \begin{pmatrix} 1 \\ -3 \end{pmatrix}$$

State which of the matrix multiplications AB , BA , CB are valid / possible and why? Then perform the matrix multiplication(s) which is/are valid (from the three stated matrix multiplications).

Show work.

☐ Need more space? Continued here: _____

$A: 3 \times 3$
 $B: 3 \times 2$
 $C: 2 \times 1$

$A \times B$ is the only valid multiplication because it is the only one that works the rules about same size row-column number. eg. $3 \times 3 \cdot 3 \times 2$

$$AB = \begin{bmatrix} 1 \cdot 0.5 + (-1)(-1) + 0 & 1 \cdot (-0.5) + (-1)(1) + 0 \\ 2 \cdot 0.5 + 0 + (-1)(1) & 2 \cdot (-0.5) + 0 + (-1)(-1) \\ -1(0.5) + 1(-1) + 1 \cdot 1 & -1(-0.5) + 1 \cdot 1 + 1 \cdot (-1) \end{bmatrix}$$

$$AB = \begin{bmatrix} \frac{3}{2} & -\frac{3}{2} \\ 0 & 0 \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

- (b) (4%) Assume that based on the three matrices above, there are three linear functions $f(x) = Ax$, $g(y) = By$, and $h(z) = Cz$. What is the matrix corresponding to the linear function that is the functional composition of the three functions $f \circ g \circ h$ (Note: order is important)?

Show work.

☐ Need more space? Continued here: _____

$f(x) = Ax$
 $g(y) = By$
 $h(z) = Cz$

$f \circ g \circ h$
 $f(g(h(z)))$

$\vec{p} = \underbrace{ABC}_{M} \cdot \vec{v}$
 Output vector M input vector

$D = BC$
 $M = AD$

BC is allowed
 AB is allowed
 ABC is allowed
 $(AB)C = A(BC) \checkmark$

3. (a) (4%) You are given the following four vectors: $v_1 = (1, -1, 2)$, $v_2 = (-1, 0.5, 1)$, $v_3 = (-2, 2, 3)$, $v_4 = (4, -4, 1)$. First, you suspect that v_4 is a superfluous vector. Thus, show whether v_4 is in $\text{Span}(v_1, v_2, v_3)$.

Show work.

☐ Need more space? Continued here: _____

$$v_1 = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} \quad v_2 = \begin{bmatrix} -1 \\ 0.5 \\ 1 \end{bmatrix} \quad v_3 = \begin{bmatrix} -2 \\ 2 \\ 3 \end{bmatrix} \quad v_4 = \begin{bmatrix} 4 \\ -4 \\ 1 \end{bmatrix}$$

$$\alpha_1 \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} + \alpha_2 \begin{bmatrix} -1 \\ 0.5 \\ 1 \end{bmatrix} + \alpha_3 \begin{bmatrix} -2 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 4 \\ -4 \\ 1 \end{bmatrix}$$

$$\begin{aligned} \alpha_1 - \alpha_2 - 2\alpha_3 &= 4 \\ -\alpha_1 + \frac{1}{2}\alpha_2 + 2\alpha_3 &= -4 \\ 2\alpha_1 + \alpha_2 + 3\alpha_3 &= 1 \end{aligned}$$

$$\begin{aligned} \alpha_1 &= 4 + \alpha_2 + 2\alpha_3 \\ -(4 + \alpha_2 + 2\alpha_3) + \frac{1}{2}\alpha_2 + 2\alpha_3 &= -4 \\ -4 - \alpha_2 - 2\alpha_3 + \frac{1}{2}\alpha_2 + 2\alpha_3 &= -4 \\ -\frac{1}{2}\alpha_2 &= 0 \\ \alpha_2 &= 0 \end{aligned}$$

$$\begin{aligned} 2(4 + 0 + 2\alpha_3) + 0 + 3\alpha_3 &= 1 \\ 8 + 4\alpha_3 + 3\alpha_3 &= 1 \\ 7\alpha_3 &= -7 \quad \alpha_3 = -1 \end{aligned}$$

$$\begin{aligned} \alpha_1 &= 2 \\ \alpha_2 &= 0 \\ \alpha_3 &= -1 \end{aligned}$$

$$2v_1 + 0v_2 + (-1)v_3 = v_4$$

$$2 \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} - \begin{bmatrix} -2 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 4 \\ -4 \\ 1 \end{bmatrix}$$

- (b) (4%) Second, now assume that you retain only v_1, v_2, v_3 . Show whether these three vectors are linearly independent.

Show work.

☐ Need more space? Continued here: _____

if the linear combination is trivial, $\rightarrow$ (all zero α) then the vectors are linearly independent

we can also try to put it on equation form and pivot around 1 to find the standard basis vectors.

$$0 = \alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3$$

$$0 = \alpha_1 \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} + \alpha_2 \begin{bmatrix} -1 \\ 0.5 \\ 1 \end{bmatrix} + \alpha_3 \begin{bmatrix} -2 \\ 2 \\ 3 \end{bmatrix}$$

we can see that none of the vectors have any thing in common except the zero vector.

when $\alpha_1 = \alpha_2 = \alpha_3 = 0$.

so they are linearly independent.

4. (a) (4%) You are given an unknown linear function $f(x) = Ax$ and only told that

$$f\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

and

$$f\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

Show what would be the output for

$$f\left(\begin{bmatrix} 3 \\ 2 \end{bmatrix}\right) = ?$$

Hint: You do not need to find A for this subtask but may use properties of linear functions. Alternatively, you may first do the subtask (b) to then solve this task.

Show work.

☐ Need more space? Continued here: _____

From the definition of linear functions, we know that $f(x+y) = f(x) + f(y)$, and that $f(cx) = cf(x)$, so

$$f\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} + 2\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = f\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) + 2f\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right)$$

$$f\left(\begin{bmatrix} 3 \\ 2 \end{bmatrix}\right) = \begin{bmatrix} 2 \\ 2 \end{bmatrix} + 2\begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2 \\ 2 \end{bmatrix} + \begin{bmatrix} 6 \\ 2 \end{bmatrix}$$

$$= \begin{bmatrix} 8 \\ 4 \end{bmatrix} //$$

- (b) (4%) State what is the size of the unknown matrix A from the task above. Also state all entries in matrix A required for the function $f(x) = Ax$ above and explain how you derived them.

Show work.

☐ Need more space? Continued here: _____

$$Ax = b$$

$$A\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

$$A\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

$$A = \begin{bmatrix} 2 & 1 \\ 2 & -1 \end{bmatrix}^{2 \times 2}$$

$$\begin{bmatrix} x_1 & x_2 \\ y_1 & y_2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} x_1 & x_2 \\ y_1 & y_2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

$$x_1 + x_2 = 3$$

$$x_1 = 2$$

$$y_1 + y_2 = 1$$

$$y_1 = 2$$

$$2 + y_2 = 1$$

check

$$\begin{bmatrix} 2 & 1 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \end{bmatrix} = 3 \cdot \begin{bmatrix} 2 \\ 2 \end{bmatrix} + 2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 6 \\ 6 \end{bmatrix} + \begin{bmatrix} 2 \\ -2 \end{bmatrix} = \begin{bmatrix} 8 \\ 4 \end{bmatrix} \checkmark$$

I used a type of a system to derive the matrix A using the $\hat{i}$ and $\hat{j}$ vectors of the transformation

5. (a) (6%) Consider the following matrix A :

$$A = \begin{pmatrix} 3 & 1 & -1 \\ 6 & 3 & 0 \\ 3 & 4 & 5 \end{pmatrix}$$

Use Gaussian Elimination (or elementary row operations) to find the echelon form matrix of matrix A . Then state explicitly what the basis vectors for A are (for A 's row space).

Show work.

☐ Need more space? Continued here: _____

$$\begin{bmatrix} 3 & 1 & -1 \\ 6 & 3 & 0 \\ 3 & 4 & 5 \end{bmatrix} \quad \begin{array}{l} R_2 \leftarrow R_2 - 2R_1 \\ R_3 \leftarrow R_3 - R_1 \end{array} \quad \begin{bmatrix} 3 & 1 & -1 \\ 0 & 1 & 2 \\ 0 & 3 & 6 \end{bmatrix} \quad R_3 \leftarrow R_3 - 3R_2$$

$\begin{bmatrix} 3 & 1 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$, the basis vectors for the row space are $v_1 = [3, 1, -1]$ & $v_2 = [0, 1, 2]$.
the third row vector is the zero vector.

- (b) (2%) Based on the results in subtask (a), could Gauss-Jordan find for matrix A its inverse A^{-1} ? Why / why not? (In case you did not complete the previous subtask, assume you obtained 3 basis vectors after Gaussian Elimination).

Show work.

☐ Need more space? Continued here: _____

No because the third vector is linearly dependent on the other vectors. You cannot derive the identity matrix from it.

6. (a) (2%) Let $v_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$, $v_2 = \begin{bmatrix} 1 \\ 7 \\ -1 \\ -7 \end{bmatrix}$ and $v_3 = \begin{bmatrix} -1 \\ 1 \\ -1 \\ 1 \end{bmatrix}$.

Determine whether the vectors v_1 , v_2 and v_3 are mutually orthogonal

Show work.

☐ Need more space? Continued here: _____

they are orthogonal if $\langle v_i, v_j \rangle = 0$.

$$\langle v_1, v_2 \rangle = 1 \cdot 1 + 1 \cdot 7 + 1 \cdot (-1) + 1 \cdot (-7) = 0 \quad \checkmark$$

$$\langle v_1, v_3 \rangle = 1 \cdot (-1) + 1 \cdot 1 + 1 \cdot (-1) + 1 \cdot 1 = 0 \quad \checkmark$$

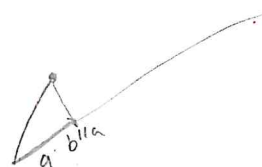
$$\langle v_2, v_3 \rangle = 1 \cdot (-1) + 7 \cdot 1 + (-1) \cdot (-1) + (-7) \cdot 1 = 0 \quad \checkmark$$

yes!

(b) (6%) Find the point p in $\text{span}\{v_1, v_2, v_3\}$ closest to $b = \begin{bmatrix} 12 \\ 0 \\ -4 \\ -12 \end{bmatrix}$

Show work.

☐ Need more space? Continued here: _____



This is a classic first engine problem. we want to find the length of the $b^{\perp v}$.

$$\|u - \text{proj}_v(u)\| = \|u^{\perp v}\|$$

$$\|b - \text{proj}_v(b)\| = \|b^{\perp v}\|$$

$$\|b\| - \|b^{\perp v}\| = \sqrt{12^2 + 0 + (-4)^2 + (-12)^2} - \|\text{proj}_v(b)\|$$

$$= 4\sqrt{19} - \frac{\langle v_1, b \rangle}{\langle b_1, b_1 \rangle} b_1 = 4\sqrt{19} - \frac{-4}{4\sqrt{19}} \cdot b_1 = \begin{bmatrix} \end{bmatrix}$$

$$v = \begin{bmatrix} 1 \\ 1 \\ 1 \\ -1 \end{bmatrix}$$

7. (8%) Consider the matrix-vector equation $Ax = b$ where A is a matrix with the singular value decomposition $A = U\Sigma V^T$ and b is a vector:

$$U = \begin{bmatrix} \frac{2}{3} & \frac{2}{3} \\ \frac{2}{3} & -\frac{1}{3} \\ \frac{1}{3} & -\frac{2}{3} \end{bmatrix}, \Sigma = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}, V^T = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, b = \begin{bmatrix} 3 \\ 3 \\ -3 \end{bmatrix},$$

Solve the matrix-vector equation $Ax = b$ by finding $\hat{x}$ and determining whether $\hat{x}$ is a valid solution.

Show work.

☐ Need more space? Continued here: _____

$$A = U\Sigma V^T$$

$$A = \begin{bmatrix} \frac{2}{3} & \frac{2}{3} \\ \frac{2}{3} & -\frac{1}{3} \\ \frac{1}{3} & -\frac{2}{3} \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} U \end{bmatrix} \begin{bmatrix} 0 & 3+0 \\ 3 & 0 \end{bmatrix}$$

$$A = \begin{bmatrix} \frac{2}{3} & \frac{2}{3} \\ \frac{2}{3} & -\frac{1}{3} \\ \frac{1}{3} & -\frac{2}{3} \end{bmatrix} \begin{bmatrix} 0 & 3 \\ 3 & 0 \end{bmatrix} = \begin{bmatrix} 0+2 & 2+0 \\ 0+2 & -1+0 \\ 0+1 & -2+0 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 2 & -1 \\ 1 & -2 \end{bmatrix}$$

$$Ax = b$$

$$\begin{bmatrix} 2 & 2 \\ 2 & -1 \\ 1 & -2 \end{bmatrix} x = \begin{bmatrix} 3 \\ 3 \\ -3 \end{bmatrix}$$

$$2x + 2y = 3$$

$$2x - y = 3$$

$$x - 2y = -3$$

$$\frac{3}{2} \neq -3$$

$$2x + 2(2x-3) = 3$$

$$2x + 4x - 6 = 3$$

$$2x - 3 = 5$$

$$6x = 9$$

$$x = \frac{9}{6} = \frac{3}{2}$$

$$2x - 3 = 5$$

$$\frac{3}{2} \cdot 2 - 3 = 0$$

$$0 = 5$$

$\hat{x}$ is not a valid.

Part 2

8. You are given the function `equal_basis` that should check whether two bases are equal, specifically, if the two have the same span. Unfortunately, the provided function is incorrect i.e. it does not provide the correct answer whether the two bases have the same span.

```
1 from solver import solve
2 from matutil import rowdict2mat
3
4
5 def equal_basis(basis1, basis2):
6
7     A = rowdict2mat(basis1)
8     for b in basis2:
9         x = solve(A, b)
10        if not (A * x - b).is_almost_zero():
11            return False
12
13    return True
```

- (a) (6%) Please explain what the function currently does and why that is not sufficient to check whether they are equal.

Currently, the function takes two basis, turns the rows of 'basis1' and turns it into a matrix. Then, it checks each vector in the second basis if the result is the zero vector. But it doesn't work because it returns false for the first vector check if the first vector isn't correct, it never moves to the next.

- (b) (6%) Please provide a modified version of the code that correctly determines whether two bases are equal or not.

☐ Need more space? Continued here: _____

```
A = rowdict2mat(basis1)
for b in basis2:
    x = solve(A, b)
    if (A * x - b).is_almost_zero():
        return True
return False
```


9. You are given two functions. The input M is a matrix where each column represents an instance (such as a senator) and each row represents a property (such as a voting result).

```
1 from vec import Vec, add, neg
2 from mat import Mat
3 from matutil import rowdict2mat, mat2rowdict, value, mat2coldict, coldict2mat, submatrix
4 from SVD import factor
5
6 def function3(M):
7     M_rows = mat2rowdict(M)
8     c = Vec(M.D[0], [(key, sum(value(row)) / len(value(row))) for key, row in M_rows])
9     return c
10
11
12 def function4(M):
13     c = function3(M)
14     M_cols = mat2coldict(M)
15     M_new_cols = {}
16     for key, col in M_cols:
17         M_new_cols[key] = add(col, neg(c))
18     M_new = coldict2mat(M_new_cols)
19
20     U, S, V = SVD.factor(M_new)
21
22     return (submatrix(U, U.D[0], U.D[1][0]),
23             submatrix(S, S.D[0][0], S.D[1][0]),
24             submatrix(V, V.D[0], V.D[1][0]))
```

- (a) (4%) Describe what the function `function3` defined above is computing. Make sure to explain the meaning of the output, not only the arithmetic operations being performed.

☐ Need more space? Continued here: _____

it looks like its creating a new vector c that has the first element as the first value in the column dict and maps the average of the rows of M .

$$c = [\text{value}, [\text{key}, \text{avg}]]$$

- (b) (8%) Describe what the function `function4` defined above is computing. Make sure to explain the meaning of the output, not only the arithmetic operations being performed. What would be the result of calling this function on the matrix of senator votes?

☐ Need more space? Continued here: _____

for each column in M , a new column is made by subtracting the c from `function3` from the original column.

And then transforming back into a matrix.

it's returning the singular value decomposition.

10. (12%) Describe what the following function foo is computing and explain what the result list contains after execution, as well as the relationship between the elements. Lastly, what would be an appropriate name for the function? Note: remember EPS is a value of (approximately) zero.

```
1 from echelon import transformation
2 from matutil import mat2rowdict
3 from vec import Vec
4
5
6 def foo(A, eps=1e-15):
7
8     M = transformation(A)
9     U = M * A
10    # U is in echelon form
11
12    rowlist = list(mat2rowdict(U).values())
13    col_labels = sorted(A.D[1], key=repr)
14
15    pivot_cols = []
16    for row in rowlist:
17        for col in col_labels:
18            if abs(row[col]) > eps:
19                pivot_cols.append(col)
20                break
21
22    free_cols = [col for col in col_labels if col not in pivot_cols]
23    result = []
24
25    for free_col in free_cols:
26        v = Vec(set(col_labels), {free_col: 1})
27        for row in reversed(rowlist):
28            for col in pivot_cols:
29                if abs(row[col]) > eps:
30                    v[col] = -row * v / row[col]
31                    break
32        result.append(v)
33
34    return result
```

This function is checking for the inverse of A.

$A^{-1} = \frac{1}{\det(A)} \cdot \text{adj}(A)$. using the RREF to pivot along diagonally. If the inverse exist that is.

A good name for the function would be

```
def find_inverse(A)
```


1. (12%) Complete the following function `orthogonalComplement`. The function takes two lists of vectors V and W that form bases for $\mathcal{V}$ and $\mathcal{W}$ respectively, such that $\mathcal{V}$ is a subspace of $\mathcal{W}$. The output is a list of vectors U that is a basis for $\mathcal{U}$, the orthogonal complement of $\mathcal{V}$ with respect to $\mathcal{W}$.

```

1 def orthogonalComplement(V, W):
2     """
3     Input:
4         V: list of Vec instances that form a basis of a vector space span(V)
5         W: list of Vec instances that form a basis of a vector space span(W)
6         such that span(V) is a subspace of span(W)
7     Output:
8         U: list of Vec instances that form a basis of a vector space span(U)
9         such that span(U) is the orthogonal complement of span(V) with regard to span(W)
10
11     Example:
12     >>> V = [Vec({0,1,2}, {0:1, 1:0, 2:0}), Vec({0,1,2}, {0:0, 1:1, 2:0})]
13     >>> W = [Vec({0,1,2}, {0:1, 1:0, 2:0}), Vec({0,1,2}, {0:0, 1:1, 2:0}), Vec({0,1,2},
14             {0:0, 1:0, 2:1})]
15     >>> U = orthogonalComplement(V, W)
16     >>> U
17     [Vec({0,1,2}, {0:0, 1:0, 2:1})]
18     """
19     pass

```

Pseudocode

well, the orthogonal complement is
 And we know that the only vector V and W
 share is the zero vector. So

$$\begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ \vdots \\ v_n \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ w_4 \\ w_n \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

```

def orthogonalComplement(v, w):
    complement_dict = {}
    for each row in V: (mat2row)
        check for complement with U columns (col2dict)
        if v*U is zero add to dict
        else: continue.
    return complement_dict

```


Part 3

2. (8%) Let V be a vector space that is spanned by the vectors v_1, v_2, v_3 , which are linearly independent. Let v_4 and v_5 be two other vectors in V . Show whether the set of vectors $\{v_1, v_2, v_3, v_4, v_5\}$ fulfills all properties to be a basis for V .

if v_1, v_2, v_3 are linearly independent, they form a basis to span V already, and according to the unique representation Lemma proof.

There is at least one representation of v in terms of basis vectors. We can write v_4 & v_5 as a linear combination of the others, so they are superfluous. We know that the vectors $\{v_1, v_2, v_3\}$ can form a basis, so adding the others to the set doesn't change the properties for being a basis for V .

13. (8%) Let $\{v_1, v_2, \dots, v_k\}$ be an orthogonal basis for $\mathcal{V}$ and $\{u_1, u_2, \dots, u_m\}$ be an orthogonal basis for $\mathcal{U}$ such that $\mathcal{V}$ is the orthogonal complement of $\mathcal{U}$ with respect to some vector space $\mathcal{W}$. Show that the only vector that is both in $\mathcal{V}$ and $\mathcal{U}$ is the zero vector.

The standard definition of the direct sum, is that the only vector in both $\mathcal{V}$ and $\mathcal{U}$ is the zero vector.

If there was any other vector in both $\mathcal{V}$ and $\mathcal{U}$, they wouldn't complement orthogonally to each other. And that is a contradiction.



4. (8%) You are given the following incorrect claim and incorrect proof concerning eigenvalues. Point out and explain the error(s) in the claim and in the proof. Then correct the claim and proof (so that the proof corresponds to the corrected claim).

Claim: The eigenvalue of the upper triangular matrix $A = \begin{bmatrix} \lambda & a & b \\ 0 & \lambda & c \\ 0 & 0 & \lambda \end{bmatrix}$ is λ^3 . $(\lambda-0)(\lambda-0)(\lambda-0) = 0$
 $\sim \lambda^3$ $\lambda^3 = 0$

Proof: By Lemma 12.3.6, a number λ^3 is an eigenvalue of A if and only if $A - \lambda^3 I$ is invertible. We get:

$$\begin{aligned} A - \lambda^3 I &= \begin{bmatrix} \lambda & a & b \\ 0 & \lambda & c \\ 0 & 0 & \lambda \end{bmatrix} - \lambda^3 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \lambda & a & b \\ 0 & \lambda & c \\ 0 & 0 & \lambda \end{bmatrix} - \begin{bmatrix} \lambda^3 & 0 & 0 \\ 0 & \lambda^3 & 0 \\ 0 & 0 & \lambda^3 \end{bmatrix} \\ &= \begin{bmatrix} \lambda & a & b \\ 0 & \lambda & c \\ 0 & 0 & \lambda \end{bmatrix} - \begin{bmatrix} \lambda^3 & 0 & 0 \\ 0 & \lambda^3 & 0 \\ 0 & 0 & \lambda^3 \end{bmatrix} \\ &= \begin{bmatrix} \lambda - \lambda^3 & a & b \\ 0 & \lambda - \lambda^3 & c \\ 0 & 0 & \lambda - \lambda^3 \end{bmatrix} \\ &= \begin{bmatrix} 0 & a & b \\ 0 & 0 & c \\ 0 & 0 & 0 \end{bmatrix} \end{aligned}$$

Wrong

which is a non-symmetric matrix. By Lemma 4.13.13, $A - \lambda^3 I$ is therefore not invertible, and so λ^3 is an eigenvalue of A . QED

$$Av = \lambda v$$

$$Av = (\lambda I)v$$

a number λ is an eigenvalue if and only if it's a scalar of a vector that doesn't get knocked off it's span after a transformation.

Non-symmetrical matrices can be inverted as long as they are a square matrix with a non-zero determinant.

Name:

Singular Value decomposition:

$$[A] = [U_1 \dots U_r] [\Sigma] \begin{bmatrix} V_1^T \\ \vdots \\ V_r^T \\ 0 \end{bmatrix}$$

U Σ V^T

V onto U

Projection: $\text{Proj}_U(v) = \frac{\langle v, u \rangle}{\langle u, u \rangle} u = VUV^T v$

Norm: $\|v\| = \sqrt{\langle v, v \rangle}$

Orthogonal: $\langle u, v \rangle = u_1 v_1 + \dots + u_n v_n = 0$

$Q_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ $Q_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$

Since these are orthogonal to each other and $Q_1^T Q_1 = I$ and $Q_2^T Q_2 = I$

Inner product: $\langle u, v \rangle = u_1 v_1 + \dots + u_n v_n$

Distance from a vector to line: $\text{dist}(u, \text{span}(V)) = \|u - \text{Proj}_U(u)\| = \|u - VUV^T u\|$

Gaussian elimination: $MA = U$

Homogeneous: $Ax = 0$

Non-Homogeneous: $Ax = b$

Row Rank = Column Rank

$\dim(V) = \text{null}(V) + \text{rank}(V)$

Rank = number of linearly independent vectors that form a basis for a space

Frobenius norm: $\|A\|_F = \sqrt{(A_{1,1})^2 + \dots + (A_{n,n})^2}$

Rank-1 approximation: $G_1 U_1 V_1^T = \tilde{A}$

For the matrix A to be Orthogonal, it has to be a square matrix

and for A to be Column-Orthogonal it has to satisfy: $A^T A = I$

S.V.D. ($A = UV^T$)

$G_1 = \|A\|_F$, $G_2 = \|A V_1\| \dots$

$\Sigma = \begin{bmatrix} G_1 & 0 & 0 \\ 0 & G_2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ V corresponding (G_1, G_2) columns in V^T

$U_1 = \frac{1}{G_1} A \cdot V_1 \dots$

U : Orthogonal basis for output space

invertible: One-to-one and one-to-one

Σ : diagonal matrix of singular values

direct sum: $\dim(u) + \dim(v) = \dim(U \oplus V)$ (only share zero vector)

V : Orthogonal basis for input space

annihilator $\dim$ theorem: $\dim(V) + \dim(V^\perp) = n$

Null space/kernel: Set of solutions for $Ax = 0 \rightarrow$ Homogeneous

Inverse: if $A \cdot B = B \cdot A = I$ or A, B are square and $A \cdot B / B \cdot A = I$ ($\det(n) \neq 0$)

Eigenvalue and Eigenvectors: $AV = \lambda V$ λ is the eigenvalue of A and V the corresponding eigenvector

QR: Q has orthogonal columns and R is upper-triangular

$R = \begin{bmatrix} \langle q_1, q_1 \rangle & \langle q_1, q_2 \rangle & \langle q_1, q_3 \rangle \\ 0 & \langle q_2, q_2 \rangle & \langle q_2, q_3 \rangle \\ 0 & 0 & \langle q_3, q_3 \rangle \end{bmatrix}$

and $Q = \frac{1}{\|a_1\|} a_1 = \frac{1}{\|a_2\|} a_2 - \text{Proj}_{Q_1}(a_2) = \frac{1}{\|a_3\|} a_3 - \text{Proj}_{Q_1}(a_3) - \text{Proj}_{Q_2}(a_3)$

Orthogonalization:

$v_1 = v_1$, $v_2 = v_2 - \text{Proj}_{Q_1}(v_2)$, $v_3 = v_3 - \text{Proj}_{Q_1}(v_3) - \text{Proj}_{Q_2}(v_3) \dots$

$\|u - w\| = \sqrt{\|u\|^2 - 2\langle u, w \rangle + \|w\|^2}$

$\|u + w\| = \sqrt{\|u\|^2 + 2\langle u, w \rangle + \|w\|^2}$

Rotation by θ : $\begin{bmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{bmatrix}$

Gaussian elimination: $MA = U$

orthogonal complement: $\begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix} \perp \begin{bmatrix} v_1 \\ \vdots \\ v_m \end{bmatrix}$

$\hat{X} = RX + O^T b$

Matrix vector multi: $\det \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix} = [1, 2] \cdot [-2, 1] = [3, 4] + [-2, 1] = [1, 3]$

$\det \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix} = [1, 2] \cdot [-2, 1] = [3, 4] + [-2, 1] = [1, 3]$

Vector matrix multi: $\begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 4 \end{bmatrix} = \begin{bmatrix} 3 & 4 & 5 \\ 4 & 5 & 6 \end{bmatrix}$

Commutator: vector v such that $Uv = 0$

if we are working with S.V.D. then we can solve $Ax = b$ by using $x = V \Sigma^{-1} U^T b$

CODE.

All to a set: $\text{all}(vname)$

Domain: S, D $D = \{x', y', z', w'\}$

Set: $VCC(\{1, 2\}, \{1:1, 2:2\})$

Max: $\text{Max}(\{1, 2\}, \{1:1\}, \{1, 2\}, \{1, 2\}, \{1, 2\}, \{1, 2\}, \{1, 2\}, \{1, 2\}, \{1, 2\}, \{1, 2\})$

```
def most-similar(sen, voting, dist):
    dist = {}
    for i in voting.keys():
        for j in sen:
            dist[i] = Policy_Compare(sen, i, voting, dist)
    highest = max(dist.values())
    for name, value in dist.items():
        if value == highest:
            return name
```

```
def find_similarity(sen, sen_ref, voting, dist):
    siml = {}
    for i in sen_ref:
        name = Policy_Compare(sen, i, voting, dist)
        siml.append(name)
    return sum(siml) / len(siml)
```

```
def is_superfluous(L, i):
    EPS = 1e-14
    if len(L) == 1:
        return L[0] == Vcc(L[0], D, {}))
    A = calculate(L[i])
    b = L[i]
    x = solve(A, b)
    residual = A * x - b
    if residual < EPS:
        return True
    else:
        return False
```

```
def is_independent(L):
    if any(is_superfluous(L, i) for i in range(len(L))):
        return False
    else:
        return True
```

```
def is_dependent(L):
    return not is_independent(L)
```

```
def in_span(L, v):
    if not L:
        return False
    new_L = L + [v]
    return is_superfluous(new_L, len(new_L) - 1)
```

```
def norm(v):
    return sum(v[i]**2 for i in range(len(v)))
```

```
def normalize(v):
    n = norm(v)
    if n < 1e-10:
        return Vcc(v, D, {}))
    return Vcc(v, D, {}))
```

VCC

```
def is_equal(A, B):
    assert A.D == B.D
    for row in A.D:
        for col in B.D:
            if A[row, col] != B[row, col]:
                return False
    return True
```

```
def add(A, B):
    assert A.D == B.D
    D = Mul(A.D, {}))
    for i in A.D:
        for j in B.D:
            A[i, j] += B[i, j]
    return A
```

```
def projection(v, u):
    factor = dot(v, u) / dot(u, u)
    return scalar.mul(factor, u)
```

Mat

```
def scalar.mul(M, u):
    assert M.D == u.D
    D = Mul(M.D, {}))
    for i in M.D:
        for j in u.D:
            M[i, j] *= u[j]
```

```
def transpose(M):
    new_matrix = Mul(M.D, {}))
    for i in M.D:
        for j in M.D:
            new_matrix[j, i] = M[i, j]
    return new_matrix
```

```
def matrix.mul(M, A, B):
    assert A.D == B.D
    D = Mul(M.D, {}))
    for i in M.D:
        for j in B.D:
            M[i, j] = sum(M[i, k] * B[k, j] for k in A.D)
```

```
def matrix.mul(M, A, B):
    assert A.D == B.D
    D = Mul(M.D, {}))
    for i in M.D:
        for j in B.D:
            M[i, j] = sum(M[i, k] * B[k, j] for k in A.D)
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    assert A.D == B.D
    D = Mul(M.D, {}))
    for i in M.D:
        for j in B.D:
            M[i, j] = sum(M[i, k] * B[k, j] for k in A.D)
```

QR tip

$A = QR$

1. Gram-Schmidt to orthogonalize columns of Q

2. $R = Q^T A$

Tips & Margins

- If it says "most similar" think dot product
- If given a transformation, look for its matrix form
- Span & can I make the standard basis?
- Homogeneous > always include the zero vector

CONVEX:

$\{x \in \mathbb{R}^n : x \in \text{conv}(S)\}$

Line segment from: $[x, y] = \{(1-t)x + ty : t \in [0, 1]\}$

CONVEX: $x(3, 2) + y(0.5, 1)$

$= x(3, 2) + (1-x)(0.5, 1)$

$= x(3, 2) + y(0.5, 1)$

$x + y = 1$

The above line segment consists of the set of points $(1-t)x + ty$ for $t \in [0, 1]$

DEFINITION:

the same thing but the scalar α, β can be negative. Resulting in an infinite line through the endpoints of the line

$\{x \in \mathbb{R}^n : x \in \text{aff}(S)\}$

The above line segment consists of the set of points $(1-t)x + ty$ for $t \in \mathbb{R}$

QR - Fact.

If a matrix is square/rectangle, it can be decomposed as: $A = QR$

A: Square/Rectangle matrix

Q: is a matrix with orthonormal columns (unit vectors, that are perpendicular to each other)

R: is a upper triangular matrix.

What we are doing:

1.) Take the columns of A: $a_1, a_2, \dots, a_n$

3.) Form Q by stacking $q_1, q_2, \dots$ as columns

4.) Form R

2.) Use G.S to construct orthonormal vectors:

$$q_1 = \frac{a_1}{\|a_1\|}$$

$$q_2 = \text{normalise}(a_2 - \text{proj}_{q_1}(a_2))$$

$$q_3 = \text{normalise}(a_3 - \text{proj}_{q_1}(a_3) - \text{proj}_{q_2}(a_3))$$

...

We have: $A = \begin{bmatrix} -1 & -1 & 1 \\ 1 & 3 & 3 \\ -1 & -1 & 5 \\ 1 & 3 & 7 \end{bmatrix}$ ' Lets construct QR so we set: $A = QR$

Step-by-step Gram-Schmidt process to get Q and R

Step 1.) compute q_1

a_1 : first column of A: $\begin{bmatrix} -1 \\ 1 \\ -1 \\ 1 \end{bmatrix}$, $\|a_1\| = \sqrt{(-1)^2 + 1^2 + (-1)^2 + 1^2} = 2$, then $q_1 = \frac{a_1}{\|a_1\|} = \begin{bmatrix} -0.5 \\ 0.5 \\ -0.5 \\ 0.5 \end{bmatrix}$, $r_{11} = \|a_1\| = 2$

Step 2.) compute q_2 (Find a vector that is orthogonal to q_1 and normalize (unit length))

$a_2 = \begin{bmatrix} -1 \\ 3 \\ -1 \\ 3 \end{bmatrix}$, $\text{proj}_{q_1}(a_2) = \frac{\langle q_1, a_2 \rangle}{\langle q_1, q_1 \rangle} \cdot q_1 = 4 \cdot q_1 = \begin{bmatrix} -2 \\ 2 \\ -2 \\ 2 \end{bmatrix}$, then $u_2 = a_2 - \text{proj}_{q_1}(a_2) = \begin{bmatrix} -1 \\ 3 \\ -1 \\ 3 \end{bmatrix} - \begin{bmatrix} -2 \\ 2 \\ -2 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$

now u_2 is orthogonal to q_1 , next normalize u_2 to get q_2 : $q_2 = \frac{u_2}{\|u_2\|} = \frac{1}{2} \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.5 \\ 0.5 \\ 0.5 \\ 0.5 \end{bmatrix}$

Step 3.) compute q_3 (Find a vector that is orthogonal to q_1 and q_2 , normalize (unit length))

$a_3 = \begin{bmatrix} 1 \\ 3 \\ 5 \\ 7 \end{bmatrix}$, $\text{proj}_{q_1}(a_3) = \frac{\langle q_1, a_3 \rangle}{\langle q_1, q_1 \rangle} \cdot q_1 = 2 \cdot q_1 = \begin{bmatrix} -1 \\ 1 \\ -1 \\ 1 \end{bmatrix}$, $\text{proj}_{q_2}(a_3) = \frac{\langle q_2, a_3 \rangle}{\langle q_2, q_2 \rangle} \cdot q_2 = 8 \cdot q_2 = \begin{bmatrix} 4 \\ 4 \\ 4 \\ 4 \end{bmatrix}$

then $u_3 = a_3 - \text{proj}_{q_1}(a_3) - \text{proj}_{q_2}(a_3) = \begin{bmatrix} 1 \\ 3 \\ 5 \\ 7 \end{bmatrix} - \begin{bmatrix} -1 \\ 1 \\ -1 \\ 1 \end{bmatrix} - \begin{bmatrix} 4 \\ 4 \\ 4 \\ 4 \end{bmatrix} = \begin{bmatrix} -2 \\ -2 \\ 0 \\ 2 \end{bmatrix}$, now normalize u_3 : $q_3 = \frac{u_3}{\|u_3\|} = \frac{1}{4} \cdot \begin{bmatrix} -2 \\ -2 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} -0.5 \\ -0.5 \\ 0 \\ 0.5 \end{bmatrix}$

then R is: $\begin{bmatrix} 2 & 0.5 & -0.5 \\ 0 & 2 & -0.5 \\ 0 & 0 & 0.5 \\ 0 & 0 & 0.5 \end{bmatrix}$

R = upper triangular with entries:

$r_{ij} = \langle q_i, a_j \rangle$ for $i \leq j$

$$R = \begin{bmatrix} \langle q_1, a_1 \rangle & \langle q_1, a_2 \rangle & \langle q_1, a_3 \rangle \\ 0 & \langle q_2, a_2 \rangle & \langle q_2, a_3 \rangle \\ 0 & 0 & \langle q_3, a_3 \rangle \end{bmatrix} = \begin{bmatrix} 2 & 4 & 2 \\ 0 & 2 & 8 \\ 0 & 0 & 4 \end{bmatrix}$$

You can also

To find R we complete: $R = A^T A$

LEMMA'S.

Unique-representation Lemma

Claim: Let $a_1, \dots, a_n$ be a basis for V. For any vector

$v \in V$, there is exactly one representation of v in terms of the basis vectors.

Proof: Let v be any vector in V. The vectors $a_1, \dots, a_n$ span V, so there

is atleast one representation of v in terms of basis vectors.

$$v = \alpha_1 a_1 + \alpha_2 a_2 + \dots + \alpha_n a_n \implies \alpha_1 a_1 + \alpha_2 a_2 + \dots + \alpha_n a_n = 0 \implies (\alpha_1 - \beta_1) a_1 + \dots + (\alpha_n - \beta_n) a_n = 0$$

but all vectors are linearly independent, so $(\alpha_1 - \beta_1) = 0$ which means

that $\alpha_1 = \beta_1$

Superfluous-Vector Lemma.

Claim: for any set S and any vector $v \in S$, if v can be written as a linear combination of the other vectors in S, then:

$$\text{Span}(S - \{v\}) = \text{Span}(S)$$

Proof: Let $S = \{v_1, \dots, v_n\}$. Suppose: $v_i = \alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_{i-1} v_{i-1} + \alpha_{i+1} v_{i+1} + \dots + \alpha_n v_n$

every vector in $\text{Span } S$ can be written as $v = \beta_1 v_1 + \beta_2 v_2 + \dots + \beta_n v_n$

$$\text{substitute } v_i: v = \beta_1 v_1 + \beta_2 v_2 + \dots + (\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_{i-1} v_{i-1} + \alpha_{i+1} v_{i+1} + \dots + \alpha_n v_n) + \beta_{i+1} v_{i+1} + \dots + \beta_n v_n$$

$$= (\beta_1 + \alpha_1) v_1 + (\beta_2 + \alpha_2) v_2 + \dots + (\beta_{i-1} + \alpha_{i-1}) v_{i-1} + (\beta_{i+1} + \alpha_{i+1}) v_{i+1} + \dots + (\beta_n + \alpha_n) v_n$$

which means that a vector can be written as a linear

combination of vectors in $S - \{v\}$

Simplified Exchange Lemma

Claim: Suppose S is a set of vectors and v is a non-zero vector in $\text{Span}(S)$ then there is a vector w in S such that

Proof: Let $S = \{v_1, \dots, v_n\}$. Since v is in the $\text{Span}(S)$ we can write

$$v = \alpha_1 v_1 + \dots + \alpha_n v_n$$

$\text{Span}(\{v\}) = \text{Span}(S)$ Since v is nonzero, atleast one of the coefficients is

$$\text{non-zero. say } \alpha_i. \text{ rewrite as: } v = \alpha_i v_i + \dots + \alpha_1 v_1 + \dots + \alpha_n v_n$$

divide by α_i to get $v = v_i + \dots$. According to Superfluous-Vector Lemma:

$$\text{Span}(S) = \text{Span}(\{v\} \cup (S - \{v_i\}))$$